

. MODERN PHYSICS – MIDTERM 1. RETAKE

Please fill your **Neptun code** and **Name** on all papers you submit. Preferably use these papers for all your calculations as well. If using any extra sheets of paper you submit, write your Neptun code and name on both sides of those too.

Give the answers with a precision of 3 significant digits, together with the units, unless otherwise specified. Note that if a problem is scored, e.g., X + Y + Z points, then *three* answers are expected. The later parts of every problem get progressively harder, make sure to start every problem.

Physical constants and formulas

$$\begin{aligned}
 c &= 3.00 \times 10^8 \text{ m/s} & h &= 6.626\,070\,15 \times 10^{-34} \text{ J s} & hc &= 1.9864 \times 10^{-25} \text{ J m} = 1239.8 \text{ eV nm} \\
 \gamma &= \frac{1}{\sqrt{1 - (v/c)^2}} & f' &= f \sqrt{\frac{1 + v/c}{1 - v/c}} & m_e c^2 &= 511 \text{ keV} \\
 \frac{e^2}{4\pi\epsilon_0} &= 2.307 \times 10^{-28} \text{ Nm}^2 & E_0 &= \frac{m_e e^4}{2\hbar^2 (4\pi\epsilon_0)^2} = 13.6 \text{ eV} & E_n &= \frac{m_e e^4}{2\hbar^2 (4\pi\epsilon_0)^2} \frac{1}{n^2} \quad (\text{hydrogen})
 \end{aligned}$$

1. Experimental verification of Special Relativity (4 points)

A muon has a mean lifetime of $t' = 2.2 \mu\text{s}$ (as measured in the reference frame *co-moving* with the muon) and it travels $\ell = 6 \text{ km}$ (as measured in the *lab's frame*) before decaying into an electron and two neutrinos. What was the speed v of the muon?

2. Electron-positron annihilation (2 + 2 + 2 + 2 + 4 = 12 points)

When an electron and a positron collide, they annihilate and emit two gamma photons: $e^- + e^+ \rightarrow \gamma + \gamma$.

Hint: Use eV units for energy and eV/c units for momentum. $m_{e^+} = m_e$.

- Observed from the center-of-mass frame, an electron comes in along the x axis with velocity $u_{e^-} = 0.4c$ and a positron with $u_{e^+} = -0.4c$. Compute the relativistic (i) kinetic energy and (ii) momentum of the two particles.
- Compute the (i) total energy of the two particles and the (ii) wavelength of the emitted photons.
- The same experiment is observed in the lab frame where the electron is at rest ($u'_{e^-} = 0$). What are the wavelengths of the two photons in the lab frame, assuming that they are emitted along the x axis?

3. Photoelectric effect in a hydrogen-like atom (2 + 2 + 4 = 8 points)

Consider Bohr's model for a Li^{++} ion ($Z = 3$). Calculate

- The energy (i) of the ground state ($n = 1$) and (ii) of the first excited state ($n = 2$).
- What is the largest wavelength λ_{max} of an incoming photon that is able to ionize the electron in state $n = 5$?

4. Rutherford scattering (3 + 3 + 3 + 3 = 12 points)

In a Rutherford scattering experiment an α particle ($Z_\alpha = 2$, $r_\alpha = 1.68 \text{ fm}$, $m_\alpha = 6.645 \times 10^{-27} \text{ kg}$) is set for collision with a gold (Au) nucleus ($Z_{\text{Au}} = 79$, $r_{\text{Au}} = 7.00 \text{ fm}$) with impact parameter $b = 0$ (i.e., the problem is one-dimensional). Assume the Au nucleus does not recoil ($m_{\text{Au}} = \infty$), and use the *non-relativistic limit*

- What is the kinetic energy of the α particle far away from the Au nucleus (i.e. at a distance $r = \infty$) so that its velocity is $v = 0$ when the two nuclei just touch, i.e., at $r = r_\alpha + r_{\text{Au}}$?
- Calculate (i) the momentum and (ii) the velocity at $r = \infty$.
- Calculate the de Broglie wavelength of the α particle at $r = \infty$.

5. Wavefunction normalization (3 + 3 + 3 + 5 = 14 points)

A particle in one spatial dimension is described by the wavefunction

$$\psi(x) = Ae^{-|x|}$$

in the interval $x \in [-\infty, \infty]$.

- Determine the constant A such that the wavefunction is normalized.
- Sketch the probability density function and calculate the probability of finding the particle in the interval (i) $0 \leq x \leq \infty$ and (ii) $0 \leq x \leq \ln(2)$.
- What is the position uncertainty Δx in this state?

Hint: the wavefunction is symmetric with respect to $x = 0$.

OFFICIAL SOLUTIONS**1. Experimental verification of Special Relativity (4 points)**

A muon has a mean lifetime of $t' = 2.2 \mu\text{s}$ (as measured in the reference frame *co-moving* with the muon) and it travels makes a track $\ell = 6 \text{ km}$ long (as measured in the *lab's frame*) before decaying into an electron and two neutrinos. What was the speed v of the muon?

Solution

In the lab frame, the lifetime of the muon is longer, due to time dilation

$$t = \frac{\ell}{v} = \gamma t' > t'.$$

Rearranging the terms and using $v = \beta c$ one obtains

$$\frac{t'}{\sqrt{1 - \beta^2}} = \frac{\ell}{\beta c}$$

and solving for β and then obtaining v yields

$$v = c \left[1 + \left(\frac{ct'}{\ell} \right)^2 \right]^{-1/2} = 0.994 c.$$

2. Electron-positron annihilation (2 + 2 + 2 + 2 + 4 = 12 points)

When an electron and a positron collide, they annihilate and emit two gamma photons: $e^- + e^+ \rightarrow \gamma + \gamma$.

- (a) Observed from the center-of-mass frame, an electron comes in along the x axis with velocity $u_{e^-} = 0.4c$ and a positron with $u_{e^+} = -0.4c$. Compute the relativistic (i) kinetic energy and (ii) momentum of both particles.
- (b) Compute the (i) total energy of the two particles and the (ii) wavelength of the emitted photons.
- (c) The same experiment is observed in the lab frame where the electron is at rest ($u'_{e^-} = 0$). (i) Compute the velocity u'_{e^+} , momentum, and total energy of the positron. (ii) What are the wavelengths of the two photons in the lab frame, assuming that they are emitted along the x axis?

Solution

- (a) The kinetic energy is given by

$$K_{e^-} = K_{e^+} = (\gamma - 1)m_e c^2 = \left(\frac{1}{\sqrt{1 - \beta^2}} - 1 \right) m_e c^2 = \left(\frac{1}{\sqrt{1 - 0.4^2}} - 1 \right) 511 \text{ keV} = 46.54 \text{ keV},$$

and the momentum is

$$p_{e^-} = \gamma m_e u_{e^-} = \frac{m_e c \beta}{\sqrt{1 - \beta^2}} = 223 \text{ keV}/c,$$

$$p_{e^+} = \gamma m_e u_{e^+} = -223 \text{ keV}/c.$$

- (b) The total energy of the particles is

$$E_{\text{tot}} = 2\gamma m_e c^2 = 1115 \text{ keV}.$$

This is converted into the energy of two photons. In the center-of-mass frame the total momentum is zero, so the two photons have opposite momenta. This means that they have the same wavelength and the energy is divided equally:

$$E_\gamma = \frac{E_{\text{tot}}}{2} = \frac{hc}{\lambda}.$$

So the wavelength is

$$\lambda = \frac{hc}{E_\gamma} = 2 \frac{hc}{E_{\text{tot}}} = \frac{2 \times 1239.8 \text{ eV nm}}{1115 \text{ keV}} = 2.224 \text{ pm}.$$

- (c) Compared to the lab frame, the previous center-of-mass frame moves at velocity $v = -u_{e^-} = -0.4c$. To calculate the wavelength of the photons in the lab frame using the result in the center-of-mass frame and the Doppler-effect formula:

$$\lambda'_\pm = \lambda \sqrt{\frac{1 \pm v/c}{1 \mp v/c}},$$

resulting in $\lambda'_+ = 3.397 \text{ pm}$ and $\lambda'_- = 1.456 \text{ pm}$. Here, the longer wavelength photon is emitted in the $+x$ and the shorter wavelength photon in the $-x$ direction.

Remark:

This is consistent with conservation of momentum, the total momentum of the two photons is

$$p'_\gamma = \frac{h}{\lambda'_+} - \frac{h}{\lambda'_-} = 364 \text{ keV}/c - 851 \text{ keV}/c = -486 \text{ keV}/c = p'_{e^+},$$

where p'_{e^+} is the momentum of the positron in the lab frame. This is calculated using the velocity addition formula, the velocity of the positron is

$$u'_{e^+} = \frac{u_{e^+} + v}{1 + \frac{v}{c^2} u_{e^+}} = -0.690c.$$

The momentum of the positron in the lab frame is

$$p'_{e^+} = \gamma' m_e u'_{e^+} = -486 \text{ keV}/c.$$

3. Photoelectric effect of Bohr's atom (2 + 2 + 4 = 8 points)

Consider Bohr's model for a Li^{++} ion ($Z = 3$). Calculate

- (a) The energy (i) of the ground state ($n = 1$) and (ii) of the first excited state ($n = 2$).
- (b) What is the largest wavelength λ_{max} of an incoming that is able to ionize the electron in state $n = 5$?

Solution

- (a) The energy of the eigenstates of the Li^{++} ion is given by

$$E_n = -Z^2 \frac{E_0}{n^2}$$

where $Z = 3$ and

$$E_0 = \frac{me^4}{2\hbar^2(4\pi\epsilon_0)^2} = 13.6 \text{ eV}.$$

Therefore

$$E_1 = -Z^2 E_0 = -122.4 \text{ eV} \quad E_2 = -Z^2 \frac{E_0}{4} = -30.6 \text{ eV}.$$

- (b) The minimal energy required to photoextract an electron in state $n = 5$ is

$$E_\gamma = hf_{\text{min}} = h \frac{c}{\lambda_{\text{max}}} = |E_5| = 1.224 \text{ eV},$$

corresponding to

$$\lambda_{\text{max}} = hc \frac{1}{|E_5|} = 253 \text{ nm}$$

4. Rutherford scattering (3 + 3 + 3 + 3 = 12 points)

Consider a Rutherford's scattering experiment, in which an α particle ($Z_\alpha = 2$, $r_\alpha = 1.68$ fm, $m_\alpha = 6.645 \times 10^{-27}$ kg) is set for collision with a gold (Au) nucleus ($Z_{\text{Au}} = 79$, $r_{\text{Au}} = 7.00$ fm) with impact parameter $b = 0$ (i.e., the problem is one-dimensional). Assume the Au nucleus has infinite mass $m_{\text{Au}} = \infty$ and does not recoil.

- What kinetic energy of the α particle at a distance $r = \infty$ from the Au nucleus, so that its velocity is $v = 0$ when the two nuclei just touch, i.e., at $r = r_\alpha + r_{\text{Au}}$?
- Calculate (i) the impulse and (ii) the velocity at $r = \infty$, in the *non-relativistic limit*.
- Calculate the corresponding de Broglie wavelength associated with the α particle at $r = \infty$.

Solution

- As the velocity is $v = 0$ at $r = r_\alpha + r_{\text{Au}}$, all the kinetic energy has transformed into potential energy

$$K = -\Delta V = -V(\infty) + V(r_\alpha + r_{\text{Au}}) = V(r_\alpha + r_{\text{Au}}) = \frac{1}{4\pi\epsilon_0} \frac{Z_\alpha Z_{\text{Au}} e^2}{r_\alpha + r_{\text{Au}}} = 4.20 \times 10^{-12} \text{ J} = 26.2 \text{ MeV}$$

obtained with $k = \frac{1}{4\pi\epsilon_0} = 8.99 \times 10^9 \text{ Nm}^2/\text{C}^2$, $e = 1.602 \times 10^{-19} \text{ C}$, and the conversion $1 \text{ J} = 6.242 \times 10^{18} \text{ eV}$.

- In the non-relativistic limit

$$K = \frac{p^2}{2m_\alpha} = \frac{1}{2}m_\alpha v^2$$

so that at $r = \infty$

$$p = \sqrt{2m_\alpha K} = 2.46 \times 10^{-19} \text{ N s}$$

and

$$v = \sqrt{\frac{2K}{m_\alpha}} = 3.56 \times 10^7 \text{ m/s},$$

where $m_\alpha = 6.645 \times 10^{-27} \text{ kg}$ was used.

- The corresponding de Broglie wavelength is given by

$$\lambda = \frac{h}{p} = 2.805 \times 10^{-15} \text{ m} = 2.805 \text{ fm},$$

where $h = 6.626 \times 10^{-34} \text{ J s}$ was used.

5. Wavefunction normalization (3 + 3 + 3 + 5 = 14 points)

A particle in one spatial dimension is described by the wavefunction

$$\psi(x) = Ae^{-|x|}$$

in the interval $x \in [-\infty, \infty]$.

- Determine the constant A such that the wavefunction is normalized.
- Sketch the probability density function and calculate the probability of finding the particle in the interval (i) $0 \leq x \leq \infty$ and (ii) $0 \leq x \leq \ln(2)$.
- What is the position uncertainty Δx in this state?

Hint: the wavefunction is symmetric with respect to $x = 0$.

Solution

- The constant is determined by the normalization condition, where it is convenient to exploit the symmetry of the wavefunction with respect to $x = 0$, so that for $x \geq 0$ one has that $|x| \rightarrow x$

$$1 = \int_{-\infty}^{\infty} dx |\psi(x)|^2 = |A|^2 \int_{-\infty}^{\infty} dx e^{-2|x|} = 2|A|^2 \int_0^{\infty} dx e^{-2x} = 2|A|^2 \left(-\frac{1}{2} [e^{-2x}]_0^{\infty} \right) = |A|^2.$$

so that

$$A = 1.$$

- Since the wavefunction is a symmetric function of x and it is normalized in the interval $x \in [-\infty, \infty]$, one has that

$$P(0 \leq x \leq \infty) = \int_0^{\infty} dx e^{-2x} = \frac{1}{2},$$

whereas

$$P(0 \leq x \leq \ln(2)) = \int_0^{\ln(2)} dx e^{-2x} = -\frac{1}{2} [e^{-2x}]_0^{\ln(2)} = \frac{1}{2} (1 - e^{-2 \ln(2)}) = \frac{1}{2} \left(1 - \frac{1}{4} \right) = \frac{3}{8},$$

where, obviously $2 \ln(2) = \ln(4)$.

- The uncertainty on the position is given by

$$\Delta x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2}.$$

Recalling that $\psi(x)$ is an eigenfunction of the position operator, one needs to calculate the following integrals

$$\langle x \rangle = \int_{-\infty}^{\infty} dx e^{-|x|} x e^{-|x|} = \int_{-\infty}^{\infty} dx x e^{-2|x|} = 0$$

which is zero by symmetry, and

$$\langle x^2 \rangle = \int_{-\infty}^{\infty} dx e^{-|x|} x^2 e^{-|x|} = \int_{-\infty}^{\infty} dx x^2 e^{-2|x|} = 2 \int_0^{\infty} dx x^2 e^{-2x} = \dots$$

which can be integrated by parts as

$$\int u dv = uv - \int v du,$$

with $x^2 = u$ and $e^{-2x} dx = dv$, thus yielding

$$\dots = \left[x^2 e^{-2x} \right]_0^{\infty} - 2 \int_0^{\infty} dx 2x \frac{1}{2} e^{-2x} = \dots$$

and the only non-zero term can be integrated by parts (again) with $x = u$ and $e^{-2x}dx = dv$, thus yielding

$$\dots = -\left[x e^{-2x}\right]_0^\infty + 2 \int_0^\infty dx \frac{1}{2} e^{-2x} = \frac{1}{2} \left[e^{-2x}\right]_0^\infty = \frac{1}{2}.$$

Finally, the uncertainty of the position operator can be evaluated as

$$\Delta x = \frac{1}{\sqrt{2}}$$